

Special Topic: Enabling Techniques and Cutting-Edge Applications of Foundation Models

RAG-leaks: difficulty-calibrated membership inference attacks on retrieval-augmented generation

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Abstract Recently, retrieval-augmented generation (RAG) systems have attracted attention for addressing issues like hallucinations and reliance on outdated knowledge in large language models (LLMs). Privacy studies have revealed that RAG systems are vulnerable to membership leakage in determining whether a specific target sample is included in the RAG knowledge base. Existing membership inference attack (MIA) methods for RAGs primarily rely on similarity scores between the system's responses and the true answers. These methods assume that a higher similarity score indicates the sample is more likely to have been used by the RAG system to enhance its response, suggesting it is a member of the knowledge base. However, this study uncovers an important insight: the similarity metric does not directly represent the membership status, instead measures the response difficulty of the sample. To address this, we propose a novel membership inference attack for RAG systems, called difficulty-calibrated membership inference attack (DC-MIA). It first classifies high-similarity samples as members, and then calibrates the membership scores of samples with comparable raw similarity scores using a likelihood ratio test. Experimental results demonstrate that our approach significantly improves the performance of membership inference attacks on RAG systems.

Keywords large language models, retrieval-augmented generation, membership inference, privacy risk, difficulty calibration

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1 Introduction

In recent years, large language models (LLMs), such as GPT-4 [1] and LLaMA [2], have garnered significant attention for their exceptional generative capabilities, achieving remarkable performance across diverse domains. Despite these advancements, LLMs still face critical challenges, including hallucinations [3] and reliance on outdated parametric knowledge [4, 5]. Retrieval-augmented generation (RAG) has emerged as a promising approach to address these limitations, enhancing LLM responses by incorporating high-quality, timely, and contextually relevant information from external knowledge bases [6, 7]. Unlike conventional fine-tuning methods, which require updating model parameters for task-specific adaptation, RAG employs a retrieval-based strategy, enabling the rapid and flexible deployment of LLMs across diverse applications. However, when applied to tasks involving sensitive information—such as generating medical radiology reports [8, 9] or providing answers to personalized queries [10, 11]—LLM-generated responses in RAG systems often rely on external knowledge bases containing private information, raising significant privacy concerns.

Studies have increasingly highlighted the significant privacy risks faced by RAG systems [12], with membership inference attacks (MIAs) [13–15] being a key threat. Traditional MIAs [16] aim to determine whether a specific data sample (referred to as the target sample) was involved in the training of a model (referred to as the target model), classifying it as a “member” if present or a “non-member” otherwise. They commonly assume that the model performs better when predicting members compared to non-members, leveraging this difference to carry out attacks [16–27]. For instance, member samples are often observed to exhibit lower loss values [17, 22–25] and entropy values [18] of output posteriors. Additionally, some researchers [16, 26, 27] trained attack models with the posteriors output by the target model to

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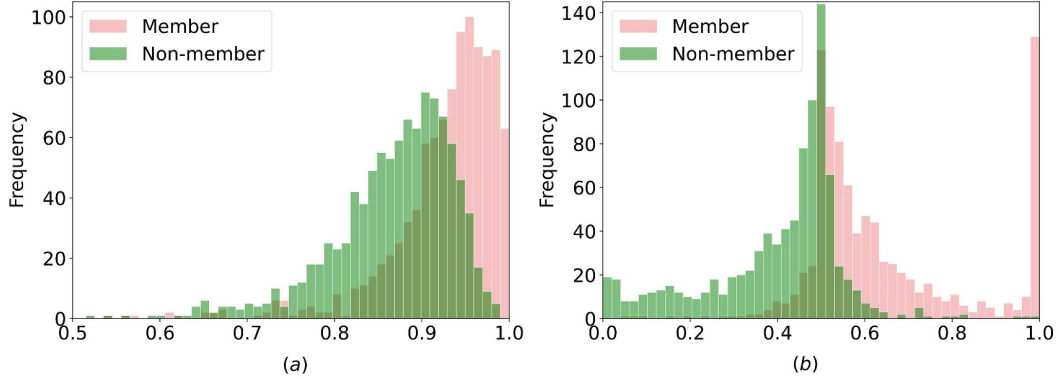


Figure 1 (Color online) Distribution of similarity scores and calibrated scores. All samples (i.e., 1000 members and 1000 non-members) are derived from the RAG system built with Mistral and AgNews. Notably, difficulty calibration substantially reduces the overlap between member and non-member distributions. (a) Similarity scores; (b) calibrated scores.

differentiate members from non-members. And a successful MIA can reveal sensitive or proprietary information. For instance, if an adversary knows that a victim's data was used to train a hospital's model, they could infer that the victim had previously visited the hospital. Furthermore, it is important to note that membership leakage in RAG systems is more severe than that caused by traditional models. This is primarily because RAG systems rely heavily on external knowledge bases for generating responses [28], which often contain more direct and precise privacy-sensitive information than the model itself.

However, the membership privacy risks in RAG systems remain inadequately explored. Anderson et al. [13] inferred membership by querying LLMs on whether the target sample appears in the context. Zou et al. [29], in contrast, demonstrated that this approach can be mitigated by prohibiting direct answers. Li et al. [14] later proposed a more practical attack by inputting the first half of a text sample into the RAG system, generating the second half, and calculating the similarity between the output and the actual second half to infer membership status. This method assumes that samples in the RAG knowledge base (i.e., members) produce outputs with higher similarity to the actual answer than non-members, but this assumption is not always reliable. As shown in Figure 1(a), the similarity scores of 1000 member and 1000 non-member samples exhibit significant overlap because what similarity measures is not membership status, but response difficulty. While membership status is a key component, response difficulty also includes sample diversity. This overlap highlights a critical limitation in existing MIAs on RAG systems, leading to an underestimation of their privacy risks. More recently, Liu et al. [15] introduced a mask-based MIA method for RAG, utilizing a masking algorithm to obscure specific words in a target sample, prompting the RAG system to predict them, and determining membership based on the accuracy of the predictions. Nonetheless, this approach also overlooks the varying levels of response difficulty across different samples.

To address this issue, we reformulate the MIA problem in RAG systems and define the concept of response difficulty formally. Then, we propose difficulty-calibrated membership inference attacks (DC-MIA), a method designed to uncover the actual underlying membership privacy leakage in RAG systems. DC-MIA builds on two key observations, as illustrated in Figure 1(a): (1) samples generating answers with exceptionally high similarity (close to 1.0) are more likely to be members, and (2) due to varying levels of difficulty across different samples, there is significant overlap in similarity scores between members and non-members (ranging from 0.5 to 0.9). Leveraging these insights, we partition the sample space into two regions: a high-similarity region, primarily containing members, and a confusion region, where similarity scores of members and non-members overlap. Samples in the high-similarity region are directly classified as members, while those in the confusion region undergo a likelihood ratio test, which evaluates whether a sample's similarity score is more likely to result from its inclusion in the RAG knowledge base or its absence. By calibrating membership probabilities in this manner, we transform raw similarity scores into likelihood ratios, referred to as calibrated scores. As depicted in Figure 1(b), this calibration substantially reduces the overlap between members and non-members within the confusion region, allowing a significant portion of member samples to become distinctly separated. Consequently, members exhibit higher likelihood ratio values, thereby improving the attack's overall performance.

We conduct extensive evaluations on multiple RAG systems constructed using three benchmark datasets and three popular LLMs, including a domestically developed Chinese LLM. Experimental results show

that DC-MIA consistently outperforms the baseline in nearly all scenarios. For example, when using Mistral with the AgNews dataset, our DC-MIA achieved a true positive rate (TPR) at 1% false positive rate (FPR) of 4.13%, compared to only 2.01% for the baseline. In terms of area under curve (AUC), our method also outperformed the baseline by nearly 16 percentage points. Moreover, we perform comprehensive ablation studies to evaluate the influence of various factors on attack performance. These factors include the number of reference RAGs, the number of retrieved items, and membership score calibration. In general, our contributions can be summarized as follows.

- We formally redefine the MIA problem in RAG systems and provide the definition of response difficulty, uncovering critical limitations in existing methods, particularly their inability to effectively address the issue of overlapping similarity scores between members and non-members.
- We propose DC-MIA, a novel two-step framework that leverages high-similarity regions recognition and likelihood ratio tests. By directly classifying samples in high-similarity regions as members and calibrating the membership probabilities of samples in confusion regions through likelihood ratio tests, DC-MIA mitigates the impact of similarity overlaps, thereby significantly enhancing attack performance.
- We conduct comprehensive experiments on multiple RAG systems built with three benchmark datasets and three popular LLMs. The results demonstrate that DC-MIA consistently outperforms baseline methods across diverse scenarios. Additionally, we carry out ablation studies to investigate the impact of various factors on attack performance.

The rest of the paper is organized as follows. Section 2 presents the related work. Section 3 defines the problem. In Section 4, we introduce our DC-MIA. Sections 5 and 6 provide an evaluation of DC-MIA across multiple RAG systems, using various benchmark datasets and LLMs. Section 7 discusses additional metrics beyond similarity and addresses the limitations of our approach. Finally, we conclude the paper in Section 8.

2 Related work

2.1 Retrieval augmented generation

RAG is a method that enhances LLMs' response quality by retrieving semantically relevant knowledge from external knowledge bases, thereby removing the reliance on computationally expensive fine-tuning [30]. This allows pre-trained LLMs to efficiently handle domain-specific generative tasks at lower costs. These advantages have significantly accelerated the practical applications of RAG technology, including personal chatbots and medical diagnostic assistants [31].

A typical RAG system comprises three key components: an LLM $\mathcal{M}$, a retriever $\mathcal{R}$, and an external knowledge base D , where the data in D follows a specific distribution $\mathbb{D}$. Each sample x from $\mathbb{D}$ consists of two parts: the query q and its corresponding ground truth answer a . Given a user query q , the retriever $\mathcal{R}$ retrieves its relevant information from D , and the model $\mathcal{M}$ generates a response a' based on the retrieved data.

Specifically, for each generation, the retriever $\mathcal{R}$'s task is to fetch the Top- k most relevant text records, denoted as $\mathcal{R}(q, D)$, from the database D in response to the user's query q . These retrieved records are then provided, along with q , to the LLM $\mathcal{M}$. Since the quality of the generated answer a' heavily depends on the relevance of the retrieved records $\mathcal{R}(q, D)$, the retriever $\mathcal{R}$ is a factor impacting a' . However, to isolate the uncertainty introduced by variations in the query q , we follow [15] and assume that the retriever $\mathcal{R}$ in the RAG system always returns the Top- k most relevant records for each query.

Subsequently, $\mathcal{M}$ generates the answer a' word by word based on $\mathcal{R}(q, D)$ and q . Let the n -th token be denoted as t_n . Following the setup in [32, 33], the answer generation process proceeds as follows:

$$t_n = \arg\max_t P_M(t|q, \mathcal{R}(q, D), t_{1:n-1}), \quad (1)$$

where $t_{1:n-1}$ represents the set of all previously generated tokens, $a' = t_{1:n}$. For simplicity, this process can be formalized as

$$a' \leftarrow M(R(q, D) || q). \quad (2)$$

Additionally, we use a similarity signal, such as the cosine similarity $S_R(a', a)$ [14], to assess the generation quality of the RAG system. Since each query q is associated with the target text x and has a unique answer a in this paper, we simplify the notation by omitting a and write it as $S_R(a')$ in the subsequent discussion.

2.2 Membership inference attack

MIAs are a common privacy threat to machine learning models. Traditional MIAs primarily focus on model-targeted attacks that aim to determine whether a given sample was used to train machine learning models [16–18, 34], federated learning models [35, 36], or LLMs [32, 37–39]. The samples are classified as members if they were included in the training set, and as non-members if they were not. Most existing MIAs are based on the observation that models tend to exhibit higher confidence when making predictions on training set members [16, 19, 20, 22]. This phenomenon is captured through a membership score, which estimates the likelihood that a sample belongs to the training set. Using this score, an adversary can either train a binary classifier or establish a threshold to effectively perform the MIA.

Moreover, many researchers [23–27, 34] have focused on improving the TPR at low FPR of an attack. Sablayrolles et al. [34] calibrated the membership score thresholds using loss values from reference models, both with and without the target sample. Watson et al. [23] introduced a simplified reference model framework, measuring how representative the sample is within the overall distribution. Carlini et al. [24] further improved this method by modeling a target sample’s loss values across reference models using Gaussian distributions. Moreover, Li et al. [27] proposed a new attack method that leverages serialized signals leaked during the training of the target model to improve the distinction between members and non-members, significantly enhancing the attack’s TPR at low FPR.

Recently, research has shown that a significant amount of private data can be extracted from LLMs in black-box settings, raising serious concerns about data privacy in LLMs, as text leakage can directly lead to privacy losses [37, 40]. For pre-trained LLMs, Duan et al. [41] evaluated the impact of pre-training data and model scale on MIA performance. Shi et al. [42] proposed an efficient min- $k\%$ detection method based on the observation that non-member texts may contain rare low-probability words. Zhang et al. [43] introduced a divergence-based calibration approach to improve the calculation of membership scores. Fu et al. [44] leveraged LLMs themselves as a more precise pre-training data detector, demonstrating strong performance on aligned LLMs and exploring potential defense mechanisms. For fine-tuned LLMs, Mireshghallah et al. [45] empirically studied how fine-tuned LLMs memorize data and assessed the feasibility of MIAs. Amit et al. [46] systematically analyzed the susceptibility of fine-tuned LLMs to MIAs and investigated various factors affecting vulnerability. Fu et al. [47] introduced a self-calibrated probabilistic variation-based MIA, which enhances MIA effectiveness by prompting the target LLM to generate datasets for fine-tuning reference models. While the privacy information in pre-trained and fine-tuned LLMs is embedded in model parameters, the private data in RAG scenarios is directly exposed in the database, posing a greater privacy threat.

For membership inference on RAGs, Anderson et al. [13] proposed an MIA method that directly queries the RAG system to determine whether the target sample appears in the context. However, Zou et al. [29] showed that this approach could be effectively mitigated with a defense prompt. Li et al. [14] later proposed a more practical attack in which the first half of a text sample is input into the RAG system, the second half is generated, and the similarity between the predicted and actual second halves is calculated to infer the text’s membership. More recently, Liu et al. [15] introduced a mask-based MIA method for RAG, where a masking algorithm is used to obscure specific words in a target sample, prompting the RAG system to predict them, and membership is determined based on the accuracy of these predictions.

3 Problem statement

3.1 Membership inference attack on RAG

Adversary’s objective. MIAs on RAG systems aim to determine whether a specific target sample x is included in the RAG system—more precisely, whether it is present in the knowledge base D used by RAG.

While MIAs have been well-defined in [16, 24], their application to RAG systems presents key differences. In traditional MIAs, the adversary seeks to infer whether a sample influenced the model’s parameters during training. However, in RAG systems, the adversary’s goal is to determine if the sample is part of the non-parametric knowledge base. As such, we propose a redefined formalization of the attack for this context, which we present below.

Definition 1 (MIA security game on RAG). The game proceeds between a challenger $\mathcal{C}$ and an adversary $\mathcal{A}$ on RAG system $\mathcal{G}$.

- (1) The challenger $\mathcal{C}$ samples a knowledge base $D \leftarrow \mathbb{D}$ for RAG system $\mathcal{G}$.
- (2) The challenger $\mathcal{C}$ flips a bit b , and if $b = 0$, samples a fresh challenge point from the distribution $\mathbb{D}$, i.e., $x \leftarrow \mathbb{D}$, such that $x \notin D$. Otherwise, the challenger selects a point from the knowledge base D , i.e., $x \leftarrow D$.
- (3) The challenger $\mathcal{C}$ sends x to the adversary $\mathcal{A}$.
- (4) The adversary $\mathcal{A}$ gets query access to the distribution $\mathbb{D}$ and the RAG system $\mathcal{G}$, and outputs a bit $\hat{b} \leftarrow \mathcal{A}^{\mathbb{D}, \mathcal{G}}(x)$.
- (5) Output 1 if $\hat{b} = b$, and 0 otherwise.

Next, we define the attack model based on the concept $\mathcal{A}$ of this security game. In this paper, we assume the adversary $\mathcal{A}$ has access to the distributions $\mathbb{D}$ and $\mathcal{G}$ (see Subsection 3.3 for details), and denote its output on x as $\mathcal{A}(x)$ for simplicity.

Attack model. Previous MIAs [16–27] primarily targeted classification models, which output posterior probability vectors. These attacks assume that the target model performs better on training set members than on non-members, i.e., it exhibits higher confidence on members. The attack model processes this confidence as a membership score (e.g., loss [22], entropy [18], modified-entropy [18], etc.) and usually uses a pre-defined threshold to predict membership $\hat{b}$. This leads to the following definition:

$$\hat{b} = I(\mathcal{A}(x) > \tau), \quad (3)$$

where I is the indicator function, τ is the membership score threshold chosen by the adversary for classification. The selection of the threshold is crucial to the performance of MIAs. Typically, it is chosen to minimize the average classification error for membership inference [18, 22].

For RAG systems, MIAs can still utilize a metric derived from the system’s responses to distinguish between members and non-members. This suggests that we can adopt (3) as the basic attack model for membership inference in RAGs, with the similarity metric proposed by [14] serving as the membership score, i.e., $S_R(a')$.

3.2 Difficulty calibration on RAG

The definition of sample difficulty is derived from classification models [23, 48–50]. It refers to how easily a model can classify a sample, typically represented by metrics such as the loss value of the predicted classification. Intuitively, if a sample is difficult to classify, the loss value is large. Carlini et al. [24] noted that sample difficulty can influence membership inference. To enhance attack accuracy, it is necessary to adjust the original membership scores based on a specific strategy, known as “difficulty calibration”. Generally, the influence of sample difficulty can be eliminated by subtracting the average membership score of the reference model from the target sample’s original membership score [23, 48, 49].

Similarly, in RAG systems, the difficulty of generating a response (denoted as response difficulty) can be described using the cosine similarity between the output of the target RAG system and the ground truth answer [13–15]. A higher difficulty in generating the answer leads to a smaller cosine similarity. Existing RAG MIA research equates response difficulty directly with sample membership to conduct membership inference. That is, samples with high cosine similarity are more likely to be members, while those with low similarity are classified as non-members. Inspired by [13, 24], we argue that, in RAG systems, sample diversity can also influence response difficulty. Relying solely on the inherent features of the samples can lead to varying cosine similarity values. To address this issue, we provide a clear definition of response difficulty in RAG systems and discuss the concept of difficulty calibration.

Definition 2 (Response difficulty). In RAG systems, the response difficulty of a text refers to how difficult it is for the RAG system to comprehend the text. To simplify the definition, we fix the LLM of the target RAG system, the response difficulty $\text{DIF}_{\text{response}}$ of a sample consists of two parts, the sample-dependent term $\text{DIF}_{\text{sample}}$ and the rag-dependent term DIF_{rag} . Formally, we define this as follows:

$$\text{DIF}_{\text{response}}(x, \mathcal{G}) = \text{DIF}_{\text{sample}}(x) + \text{DIF}_{\text{rag}}(x, \mathcal{G}). \quad (4)$$

For a fixed LLM M , $\text{DIF}_{\text{sample}}$ (depends on x) reflects the difficulty of answering x only depends on the knowledge embedded in M , specifically knowledge derived from the training set. This part is determined solely by the sample’s intrinsic characteristics and is independent of whether x is part of the RAG knowledge base. The second part DIF_{rag} (depends on $x, \mathcal{G}$) is influenced by the membership of x , that is, whether x exists in the RAG knowledge base. If x is in the RAG knowledge base, the small

DIF_{rag} results in a lower overall response difficulty $\text{DIF}_{\text{response}}$. In short, $\text{DIF}_{\text{sample}}$ corresponds to the sample diversity, while DIF_{rag} corresponds to the true membership status.

Based on this definition, we highlight that existing studies have not considered sample diversity and directly equated cosine similarity (overall response difficulty) with membership status, which is unreasonable. Hence, we introduce a difficulty calibration attack method that aligns the sample diversity between members and non-members, removing the impact of $\text{DIF}_{\text{sample}}$ on response difficulty in (4), while retaining DIF_{rag} (membership status) for classification. In other words, we consider both sample diversity and membership status as factors influencing response difficulty and aim to eliminate the misleading impact of sample diversity on response difficulty. We adopt a baseline construction approach, ensuring that both baselines include the same sample diversity $\text{DIF}_{\text{sample}}$ for comparison. Specifically, we construct response difficulty baselines for x both when it is in the RAG knowledge base (inRAG) and when it is not (outRAG), and use likelihood ratio testing to determine which scenario is more likely for sample x . As a result, the likelihood ratio serves as the calibrated difficulty value, which corresponds to the membership status.

3.3 Threat model

In this paper, we focus on the most challenging and realistic black-box scenario, where the adversary only has black-box access to the target RAG system. Specifically, the scenario is defined as follows.

Access capability. Given a target sample x , the adversary can only submit x 's query q to the RAG, and observe the generated answer a' (i.e., the textual output), but has no access to the tokenizer, probability values, or the retrieval results $\mathcal{R}(q, D)$ produced by the retriever $\mathcal{R}$.

Knowledge of target RAG. The adversary is assumed to have knowledge of the structure and parameters of the RAG system, including detailed information about both the LLM and the retriever components. Consequently, the adversary is capable of downloading the identical LLM and retriever from public sources, thereby reconstructing a replica of the target RAG system, with the exception of the knowledge base D .

Knowledge of data. The adversary is assumed to be aware of the distribution $\mathbb{D}$ that governs the target RAG knowledge base and is capable of sampling data from this distribution. Furthermore, the adversary holds a small number of member samples from the target RAG knowledge base, which together form an auxiliary dataset. The existence of such an auxiliary dataset is also assumed in existing RAG MIAs [14, 24].

4 Attack methodology

4.1 Intuition

As aforementioned, different samples exhibit varying levels of response difficulty, caused by the joint effect of two factors, sample diversity and membership status. A sample with a relatively low similarity score may either have a low $\text{DIF}_{\text{sample}}$ or be a member, resulting in a lower DIF_{rag} . As shown in Figure 1(a), there is a significant overlap between member and non-member samples in the similarity range of 0.5 to 0.9. This suggests that easier questions exhibit high similarity not because they are members, but their $\text{DIF}_{\text{sample}}$ is lower, reducing response difficulty. In other words, sample diversity is a non-ignorable factor affecting response difficulty, which interferes membership classification. Before leveraging similarity for membership inference, the impact of sample diversity must be calibrated. Therefore, existing MIA methods for RAG, which rely on similarity as a membership score, are inherently flawed. This leads to limited attack performance and underestimates the privacy risks of RAG leakage. These observations motivate us to explore the underlying causes of this issue.

Intuition 1. Firstly, we aim to use similarity to distinguish samples with clear query and membership, leaving those with ambiguous membership for later processing. In short, Phase 1 recognizes samples with both low $\text{DIF}_{\text{sample}}$ and low DIF_{rag} , which correspond to simple members. This means that only samples with both difficulty components at low levels have the highest similarity. As shown in Figure 1(a), in the region of very high similarity (close to 1.0), non-members are rare, while in the low similarity range (0.5 to 0.9), both members and non-members are mixed. This phenomenon arises from various factors influencing the similarity of a RAG system's response, including the LLM's ability to answer the sample, whether the sample is in the RAG knowledge base (i.e., membership), the retriever's performance, and

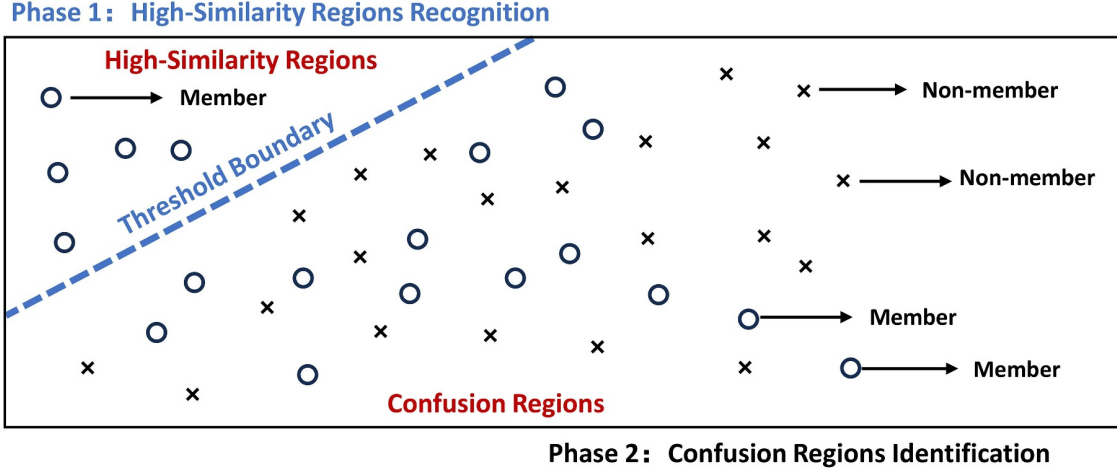


Figure 2 (Color online) Illustration of sample space segmentation of DC-MIA.

others. Any missing factor could prevent the RAG from generating a response that is highly similar to the ground truth. Therefore, samples with very high similarity are likely members, while low similarity responses may stem from any of these factors and cannot be immediately classified as non-members. Based on this, our first intuition is to classify samples with high similarity, above a certain threshold, as members, as shown in Figure 2.

Intuition 2. The more challenging problem is distinguishing between members and non-members with similar similarity scores, especially in the regions shown in Figure 1(a) (excluding the extremely high similarity region). In short, Phase 2 aims to identify between samples with low DIF_{sample} and high DIF_{rag} , and samples with high DIF_{sample} and low DIF_{rag} , which may cause confusion. Since a RAG system's similarity score is influenced by multiple factors, the key question is whether we can isolate the membership while holding other factors constant. To address this, we introduce difficulty calibration, where factors unrelated to membership are treated as the “difficulty” of the sample's response and are fixed. Specifically, we construct two reference RAG systems for the target sample: one that includes the sample (inRAG) and one that does not (outRAG). By inputting the target sample into both systems, we obtain two reference similarity scores, and their difference is theoretically determined solely by the membership. The actual similarity score from the target RAG system is then compared to these reference scores: if it is closer to inRAG, the sample is classified as a member; otherwise, it is classified as a non-member.

In summary, as shown in Figure 2, our first intuition is to isolate the high similarity score region and classify those samples as members. The second intuition uses reference RAGs and incorporates sample difficulty to design a calibrated membership score, improving the accuracy of membership inference in the confusion region.

4.2 Overview of DC-MIA

As shown in Figure 3, the adversary holds a target sample x , which includes a query q and the ground truth answer a . The adversary can send the query q to the target RAG system, just like a normal user, to receive the response a' . Then, locally and without being detected by the target RAG, the adversary can perform our DC-MIA on the target sample x , which involves two phases: high-similarity regions recognition and confusion regions identification.

High-similarity regions recognition. As described in Intuition 1, some samples receive very high similarity scores in their responses from the target RAG, indicating they are likely member samples. In this phase, the adversary uses an auxiliary dataset, composed of a few samples with known membership labels, to obtain responses from the target RAG system, and calculate similarity scores with the ground truth answers. A threshold is then identified to separate the samples into two regions: one containing members with extremely high similarity (denoted as high-similarity regions), and the other consisting of a mix of members and non-members (denoted as confusion regions). Finally, the adversary applies this threshold to the target sample: if its similarity score exceeds the threshold, it is classified as a member; otherwise, it is passed to the second phase of the attack.

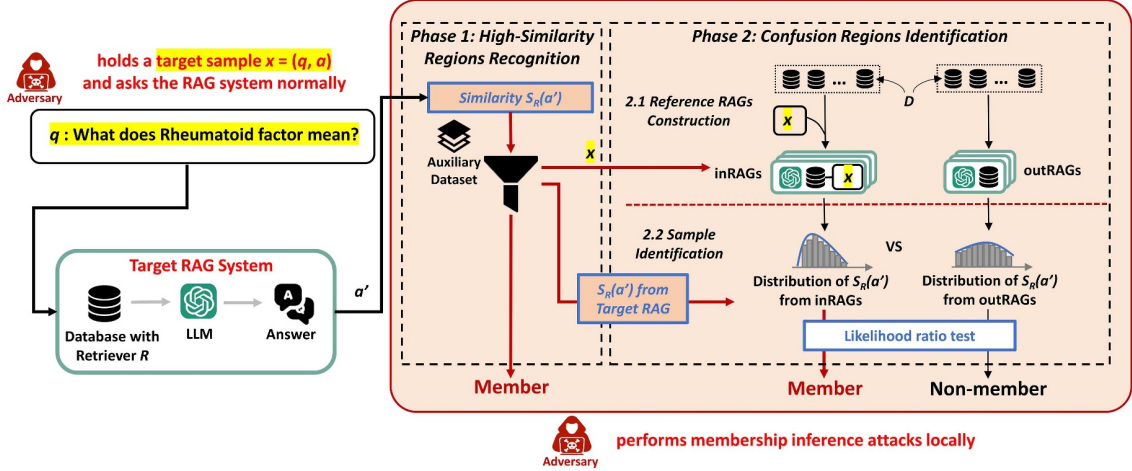


Figure 3 (Color online) Overview. DC-MIA performs a two-phase, difficulty-calibrated MIA on a given target text. The white area represents normal access to the target RAG systems, while the pale red area highlights the membership inference process.

Confusion regions identification. This attack phase aims to distinguish between members and non-members with comparable similarity scores in the confusion regions. According to Subsection 3.3, the adversary knows the structure and parameters of the target RAG system, enabling them to construct two sets of reference RAG systems with the same settings: inRAGs (with the target text in their knowledge base) and outRAGs (without it). The similarity scores of the target sample in these systems are used as reference values, which are then compared with the actual similarity score from the target RAG to infer the sample's membership.

Note that multiple inRAGs and outRAGs are used to account for random factors that affect RAG system responses. For example, the retriever may occasionally fail to fetch relevant content, and the LLM may occasionally not utilize contextual information. Using multiple inRAGs and outRAGs yields more stable inferences than relying on just one of each.

4.3 Phase 1: high-similarity regions recognition

In the first phase, our main task is to determine a threshold τ_1 for identifying high-similarity regions. To do this, we input the samples from the auxiliary dataset into the target RAG and obtain their similarity scores. Since we know the ground truth membership relationships of these samples, we can determine a threshold that optimizes the inference performance on them, as defined by (3). However, unlike existing MIA approaches targeting RAGs [13–15], our goal here is not to maximize the accuracy of membership inference. Instead, we focus on maximizing the TPR while maintaining a low FPR. For example, we might choose the threshold that corresponds to the maximum TPR at 1% FPR. This ensures that the high-similarity regions contain only a small number of non-members, while classifying as many members as possible directly into that region. Based on the selected threshold, we then partition the sample space using (5). If $\hat{b} = 1$, the sample is classified into the high-similarity region and directly identified as a member. If $\hat{b} = 0$, the sample is assigned to the confusion region and proceeds to the second phase.

$$\hat{b} = I(S_R(a') > \tau_1). \quad (5)$$

Moreover, Eq. (5) separates a subset of the member region from the overall MIA task, thereby narrowing the sample space of the confusion region. This focuses the adversary's attention and enhances attack efficiency for the next phase.

4.4 Phase 2: confusion regions identification

The second phase focuses on performing membership inference on target samples from the confusion regions, which includes both members and non-members. This phase is further divided into two steps: reference RAGs construction and sample identification.

Reference RAGs construction. As outlined in the threat model in Subsection 3.3, the adversary knows the data distribution $\mathbb{D}$ of the target RAG's knowledge base. Therefore, we randomly sample m

independent and identically distributed (i.i.d.) datasets from $\mathbb{D}$ to create m knowledge bases. These knowledge bases do not contain the target sample x , and are denoted as $D_1^{\text{out}}, D_2^{\text{out}}, \dots, D_m^{\text{out}}$. We then add the target sample x to each of these m knowledge bases, resulting in m knowledge bases that contain the target sample, denoted as $D_1^{\text{in}}, D_2^{\text{in}}, \dots, D_m^{\text{in}}$. These knowledge bases are used to construct m pairs of inRAG/outRAG systems locally, as depicted in Figure 3. It is important to note that the purpose of constructing m pairs of inRAG/outRAG systems is to prevent the simulation bias caused by constructing only a single RAG system. To sum up, the knowledge bases of inRAGs are randomly sampled from $\mathbb{D}$ and additionally include the target sample x (i.e., $D_1^{\text{in}}, D_2^{\text{in}}, \dots, D_m^{\text{in}}$), while the knowledge bases of outRAGs exclude x (i.e., $D_1^{\text{out}}, D_2^{\text{out}}, \dots, D_m^{\text{out}}$).

Sample identification. According to the security game defined in Subsection 3.1, for a given target sample $x = (q, a)$, our attack aims to distinguish whether the response a' from the target RAG $\mathcal{G}$ originates from the inRAGs or outRAGs. We denote x 's output distributions as $Q_{\text{in}}(x) = \{a' \leftarrow M(R(q, D \cup \{x\}) \| q) \mid D \leftarrow \mathbb{D}\}$ and $Q_{\text{out}}(x) = \{a' \leftarrow M(R(q, D) \| q) \mid D \leftarrow \mathbb{D}\}$, respectively. Thus, the MIA on RAGs can be formulated as a hypothesis test,

$$\Lambda(x) = \frac{p(a' | Q_{\text{in}}(x))}{p(a' | Q_{\text{out}}(x))}. \quad (6)$$

However, the above two distributions $Q_{\text{in}}(x)$ and $Q_{\text{out}}(x)$ are intractable to compute directly. To simplify this situation, we instead introduce the aforementioned similarity $S_R(a')$ to build two alternative distributions, i.e., $\tilde{Q}_{\text{in}}(x) = \{S_R(a') | M(R(q, D \cup \{x\}) \| q), D \leftarrow \mathbb{D}\}$ as the distribution of similarity of inRAGs and then $\tilde{Q}_{\text{out}}(x) = \{S_R(a') | M(R(q, D) \| q), D \leftarrow \mathbb{D}\}$ for outRAGs. Consequently, given a target sample x and the target RAG's response a' , our goal is to determine whether the similarity $S_R(a')$ is drawn from the distribution $\tilde{Q}_{\text{in}}(x)$ or $\tilde{Q}_{\text{out}}(x)$.

Fixing the Type I error probability (FPR), the Neyman-Pearson lemma [51] provides the optimal testing method. This is achieved by thresholding the likelihood ratio between the two hypotheses, enabling a test that minimizes the FPR while simultaneously maximizing the TPR. Therefore, the likelihood ratio in Figure 3 can be calculated as follows:

$$\Lambda(x) \approx \frac{p(S_R(a') | \tilde{Q}_{\text{in}}(x))}{p(S_R(a') | \tilde{Q}_{\text{out}}(x))}, \quad (7)$$

where $p(S_R(a') | \tilde{Q}_{\text{in/out}})$ represents the probability density functions of the similarity $S_R(a')$ under its fixed distributions $\tilde{Q}_{\text{in/out}}$. This is a likelihood ratio test for a one-dimensional statistic, where $S_R(a')$ can be computed by querying the target RAG system. To estimate the above signal, we model each of $\tilde{Q}_{\text{in/out}}$ as a parameterized Gaussian distribution. For $\tilde{Q}_{\text{in}}(x)$, we input the x 's query q to m inRAGs, obtaining m similarities. The distribution $\tilde{Q}_{\text{in}}(x)$ will be fitted as a univariate Gaussian distribution N_{in} with parameters μ_{in} and σ_{in} derived from the m similarities. Similarly, $\tilde{Q}_{\text{out}}(x) = N_{\text{out}}$. Finally, x 's sample-dependent signal $\Lambda(x)$ (i.e., calibrated score) is calculated by plugging the actual similarity $S_R(a')$ into (7).

Subsequently, we construct an attack model in the following form:

$$\hat{b} = I(\Lambda(x) > \tau_2). \quad (8)$$

Specifically, we determine the threshold τ_2 for the calibrated score $\Lambda(x)$ by selecting it to maximize a chosen evaluation metric, such as AUC or TPR at a fixed low FPR, across the auxiliary dataset. Moreover, for each sample in the confusion region, $\Lambda(x)$ is calculated using (7), while high-similarity samples, already classified as members in Phase 1, are assigned the maximum value of 1. As shown in Figure 1(b), after the above calibration, the overlap between member and non-member distributions is significantly reduced.

5 Experimental setup

5.1 Datasets

We consider three distinct benchmark datasets to conduct our experiments.

- **HealthCareMagic**¹⁾. HealthCareMagic is a question-answering dataset focused on medical dialogues, containing approximately 110000 data points.
- **AgNews**²⁾. AgNews is a text classification dataset, which consists of 120000 news articles. In our experiments, we focus solely on the text content, excluding the category labels. Following [14, 32], we split each article into two parts: the first part serves as the query for the RAG system, while the second part is treated as the ground truth response.
- **NaturalQuestions**³⁾. Furthermore, in Subsection 7.3, we also employed another dataset, natural questions (NQ), to verify the effectiveness of the DC-MIA attack. This is a large-scale question-answering dataset that covers a vast number of questions and answers across various domains and is widely used in the field of RAG research [29, 52].

Specifically, we randomly selected 8000 samples from each dataset to construct the target RAG system, where these 8000 samples are considered as members, and the remaining samples are treated as non-members. To create a balanced evaluation dataset, we randomly selected 1000 members from the 8000 members and 1000 non-members from the remaining non-members, respectively. In addition, following [14], we made an additional random selection of 100 members and 100 non-members to serve as an adversary-known auxiliary dataset. Finally, when constructing the reference RAG systems, we randomly sampled from the remaining non-members, maintaining each knowledge base of 8000 samples. If the samples of the evaluation dataset (i.e., target samples) are further added in a reference RAG system, the system is termed inRAG; otherwise, it is referred to as outRAG.

Note that in [14], the evaluation and auxiliary datasets are of equal size (200 samples each), which assumes the adversary has access to an auxiliary dataset of the same size as the evaluation dataset, potentially biasing the results. In contrast, we use a 10 : 1 ratio between the evaluation and auxiliary datasets, which is more reasonable.

5.2 RAG systems

Model and retriever. In our experiments, we utilized two popular LLMs, LLaMA [2] and Mistral [53], along with a domestically developed model, ChatGLM [54]. Given that the objective of our experiments is to investigate privacy leakage in RAG systems rather than in LLMs, we selected the most common models with parameter sizes ranging from 7B to 9B, instead of opting for larger ones. Specifically, we used Meta-Llama-3-8B-Instruct (8B parameters) for instruction tasks, Mistral-7B-Instruct-V02 (7B parameters) for following instructions, and glm-4-9b-chat (9B parameters) for generating conversational responses. All of these models are open source and available for download from the Internet. And for the retriever, following [14, 15], we used all-MiniLM-L6-v2 [55] to embed both the queries and the texts in the knowledge base, and employed FAISS [56] for storing and searching the texts.

Prompt template. The RAG prompt template is shown below:

```
Please answer the question based on the provided context.
Context: {context}
Question: {user_prompt}
```

where {context} refers to the Top- k records retrieved from the knowledge base, with k defaulting to 4. In the ablation study, we further investigated the impact of the Top- k parameter.

Moreover, for the HealthCareMagic dataset, we directly put the question into {user_prompt}. In contrast, for AgNews, we truncate the first ten words of each sample and place them in the {text} field of the following {user_prompt}.

```
Complete this sentence: {text} based on the context.
```

Finally, both the target RAG and reference RAGs have the same structure as described above, with the number of reference RAGs set to 16 by default in our experiments (i.e., 8 inRAGs and 8 outRAGs).

5.3 Baseline

To demonstrate the effectiveness of our DC-MIA, we compared it with the baseline S-MIA proposed by Li et al. [14], which assumes that a sample belonging to the RAG system will exhibit significant similarity

1) <https://huggingface.co/datasets/lavita/ChatDoctor-HealthCareMagic-100k>.

2) https://huggingface.co/datasets/fancyzhx/ag_news.

3) https://huggingface.co/datasets/google-research-datasets/natural_questions.

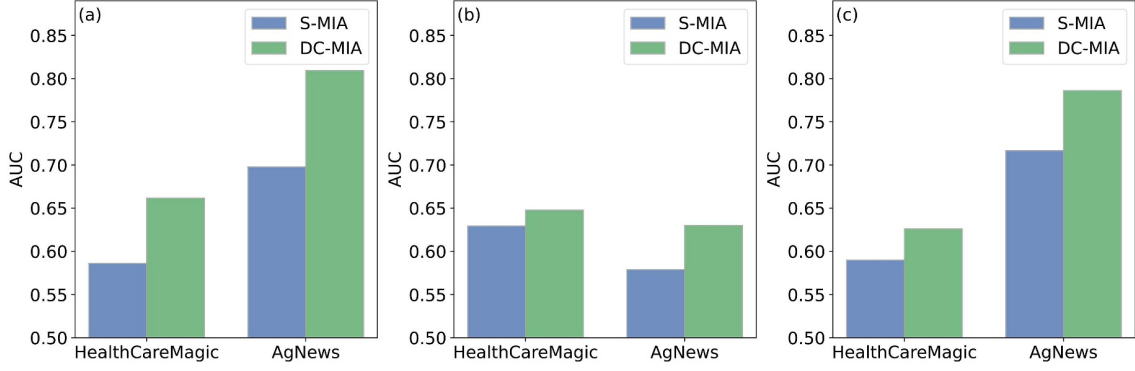


Figure 4 (Color online) Comparison of DC-MIA and S-MIA attacks on RAG systems in terms of AUC, using different LLMs and datasets. (a) Mistral-based RAGs on AgNews and HealthCareMagic; (b) LLaMA-based RAGs on AgNews and HealthCareMagic; (c) ChatGLM-based RAGs on AgNews and HealthCareMagic.

to the RAG-generated response. They split each sample into two parts: the first half is used as a query to RAG, and the second half as the ground truth answer. The similarity between the RAG response and the ground truth is then used as a membership score. While they also consider perplexity as a membership score, it cannot be obtained in the black-box setting we focus on. Therefore, we use only similarity as the membership score for the baseline.

Note that the MIA methods on RAG proposed in [13, 15] are not included as baselines for comparison, for the following reasons.

Of limited practical relevance. Anderson et al. [13] proposed an MIA method for RAG by directly querying the RAG system whether the target sample appears in the context. However, Zou et al. [29] showed that this approach can be effectively mitigated by prohibiting direct answers in the prompt, which is why we do not include it as a baseline.

Source code unavailable. Concurrently, Liu et al. [15] proposed a mask-based MIA, which uses a masking algorithm to conceal words in the target sample before inputting it into the RAG system. The attack then leverages the accuracy of mask predictions from the RAG’s output as the membership score. Since its code is not publicly available, we attempted to reproduce the attack but were unsuccessful. As a result, we are unable to compare our approach with it. It is worth noting that this method still ignores sample difficulty variations, limiting its attack performance. Therefore, our DC-MIA could theoretically enhance its effectiveness.

5.4 Evaluation metrics

In our experiments, we adopt two popular evaluation metrics in existing MIAs [16–27, 48].

AUC. AUC is the area under the receiver operating characteristic (ROC) curve, used to measure the separability between members and non-members. It is the most commonly used metric in existing works.

TPR at 1% FPR. TPR at 1% FPR, proposed by [24], focuses on an attack’s true positive rate at low false positive rates. This is because an attack with a high TPR at a high FPR is unreliable for the adversary and tends to have low value. As a result, it has been widely used in [25–27, 48].

6 Experimental results

6.1 Attack performance

The attack performance of our DC-MIA and the baseline is illustrated in Figures 4 and 5.

First, we observe that DC-MIA consistently achieves optimal performance across nearly all scenarios. For example, on the Mistral with HealthCareMagic dataset, our DC-MIA achieved over 4% TPR at 1% FPR, whereas S-MIA reached less than 1.5%. As for AUC, our method achieved 0.662, compared to 0.586 for S-MIA.

This superior performance can be attributed to our focus on the large overlap in similarity scores between members and non-members, as illustrated in Figure 1(a). While members of the RAG typically have higher similarity, some non-members that are easier for the RAG to answer may have similarity scores

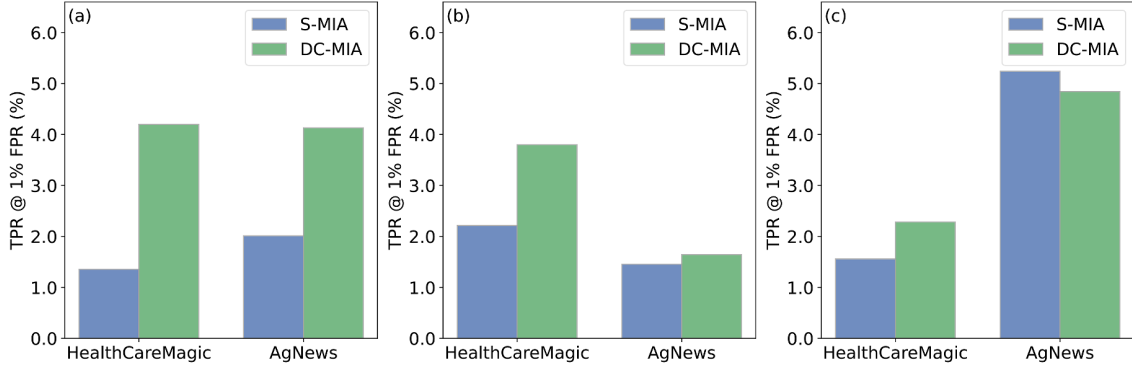


Figure 5 (Color online) Comparison of DC-MIA and S-MIA attacks on RAG systems in terms of TPR at 1% FPR, using different LLMs and datasets. (a) Mistral-based RAGs on AgNews and HealthCareMagic; (b) LLaMA-based RAGs on AgNews and HealthCareMagic; (c) ChatGLM-based RAGs on AgNews and HealthCareMagic.

Table 1 Impact of Phase 2 for different RAGs.

Attack method	HealthCareMagic					
	AUC			TPR at 1% FPR (%)		
	Mistral	LLaMA	ChatGLM	Mistral	LLaMA	ChatGLM
DC-MIA	0.661	0.648	0.626	4.198	3.800	2.282
Phase-2-only	0.653	0.631	0.622	3.942	3.776	1.811
Attack method	AgNews					
	AUC			TPR at 1% FPR (%)		
	Mistral	LLaMA	ChatGLM	Mistral	LLaMA	ChatGLM
DC-MIA	0.809	0.630	0.786	4.129	1.643	4.843
Phase-2-only	0.800	0.622	0.782	3.288	1.520	4.098

close to those of members. Conversely, some members that are harder to answer may have similarity scores resembling those of non-members. As a result, S-MIA, which relies on raw similarity as the membership score, struggles to accurately classify these “hard” members and “easy” non-members. In contrast, DC-MIA employs a two-phase attack framework. In the first phase, we identify members with high raw membership scores (similarity), and in the second phase, we apply a likelihood ratio test to transform the membership score of a sample from raw similarity to a calibrated score. This significantly enhances the distinction between members and non-members, as shown in Figure 1(b).

Additionally, we found that the privacy risk in RAG systems varies depending on the LLM used. The privacy risk associated with LLaMA is lower than that of the other two models. Moreover, different datasets used as knowledge bases in RAG systems exhibit varying levels of risk. Interestingly, the domestically developed large model, ChatGLM, results in the least privacy leakage when paired with HealthCareMagic. Even with our attack method, it is challenging to achieve strong attack performance in this scenario. However, when ChatGLM is combined with AgNews, it exhibits a significant privacy risk. The underlying reasons for this phenomenon warrant further analysis in the future. In summary, our experimental results expose the true privacy risks in RAG systems, highlighting a critical issue that previous studies have overlooked: they have significantly underestimated these risks by failing to account for the varying difficulty levels across different samples.

6.2 Ablation study

In this subsection, we examine the impact of several key factors on the attack performance of our method.

Impact of membership score calibration (Phase 2). As described in Subsection 4.2, our approach consists of two phases. The second phase is the most challenging and crucial, as it focuses on distinguishing between members and non-members when their similarity scores are comparable (i.e., samples in the confusion region). To explore this further, we tested an attack that involves only the second phase, applying the likelihood ratio test to all samples, rather than limiting it to those in the confusion region. This method is referred to as “Phase-2-only”. As shown in Table 1, the “Phase-2-only” attack still outperforms the baseline in most cases. For example, on Mistral with HealthCareMagic, “Phase-2-only” achieves a TPR of 3.942% at 1% FPR, while the baseline performs below 1.5% (see Figure 5). This

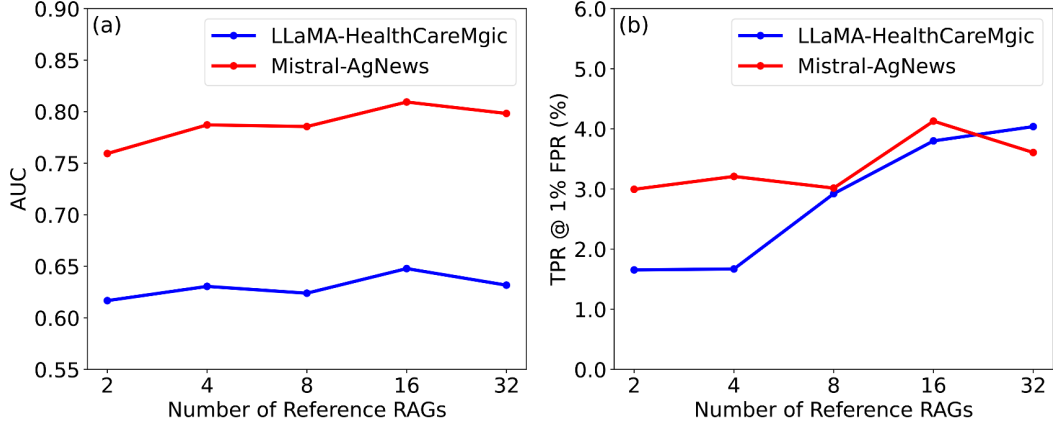


Figure 6 (Color online) Impact of the number of reference RAGs: experiments on LLaMA with HealthCareMagic and Mistral with AgNews. (a) AUC variation with respect to the number of reference RAGs; (b) TPR at 1% FPR variation with respect to the number of reference RAGs.

underscores the importance of considering the impact of sample difficulty on similarity and calibrating the membership score accordingly. Additionally, we observe that the first phase of our attack can further improve attack performance, reinforcing our observation that samples with very high similarity are likely to be members.

Number of reference RAGs. The number of reference RAGs used in our attack affects both the accuracy of modeling the Gaussian distribution and the adversary’s computational cost. Therefore, determining the optimal number of reference RAGs is critical. Figure 6 illustrates the impact of the number of reference RAGs on attack performance. We observe that increasing the number of reference RAGs improves attack performance. However, performance peaks at 16 reference RAGs, and increasing the number beyond this point typically results in a decline. Therefore, the computational cost of executing our attack is limited, with optimal performance typically achieved with around 16 reference RAGs.

Number of retrieved items (Top- k). Figure 7 illustrates the impact of the number of retrieved items by the RAG retriever on attack performance. It is observed that our DC-MIA attack method significantly outperforms the baseline in most cases, regardless of the number of retrieved items. As the number of retrieved items increases, the attack performance of both DC-MIA and the baseline generally decreases, though occasional fluctuations are observed. For example, in the ChatGLM scenario, the baseline exhibits a higher TPR at 1% FPR for Top- $k = 6$ compared to Top- $k = 1$, yet the AUC still decreases. These findings are consistent with the overall trend previously discussed. This suggests that when the RAG system uses more context to answer questions, it diminishes its focus on the most relevant information. Furthermore, when only one item is retrieved, the AUC of our attack nearly reaches 0.9, with the TPR at 1% FPR reaching 8%. This indicates that if the RAG system retrieves only the most relevant item for each query, the risk of content leakage is significant. Therefore, we recommend increasing the number of retrieved items while maintaining RAG system performance to reduce privacy risk.

Impact of retriever. To test the robustness of DC-MIA in RAG systems across different retrieval methods, we evaluated its performance using two additional retrievers: BGE-en [57] (dense vector-based retrieval) and BM25 [58] (sparse retrieval). As shown in Table 2, DC-MIA outperforms the baseline method, S-MIA, in both AUC and TPR at 1% FPR. For example, with the AgNews dataset and the BGE retriever, DC-MIA achieved an AUC of 0.782 and a TPR of 3.948% at 1% FPR, while S-MIA reached only 0.675 and 2.340%, respectively. This demonstrates both the effectiveness and adaptability of the DC-MIA framework across different retrieval methods. Furthermore, we conducted a controlled experiment under an ideal retrieval scenario, assuming that the target sample was present in the vector database with 100% recall (the theoretical upper bound of retriever performance). The results in Table 2 show that DC-MIA performs significantly better in this ideal scenario compared to real-world retrieval systems. This underscores the strong relationship between retrieval quality and attack effectiveness, as the DC-MIA mechanism relies on the differences between member and non-member retrieval results. Higher retrieval recall leads to more pronounced differences, thereby enhancing the effectiveness of the attack. In conclusion, these findings demonstrate that DC-MIA exhibits robust attack performance against various retrievers, including both ideal and real-world retrievers.

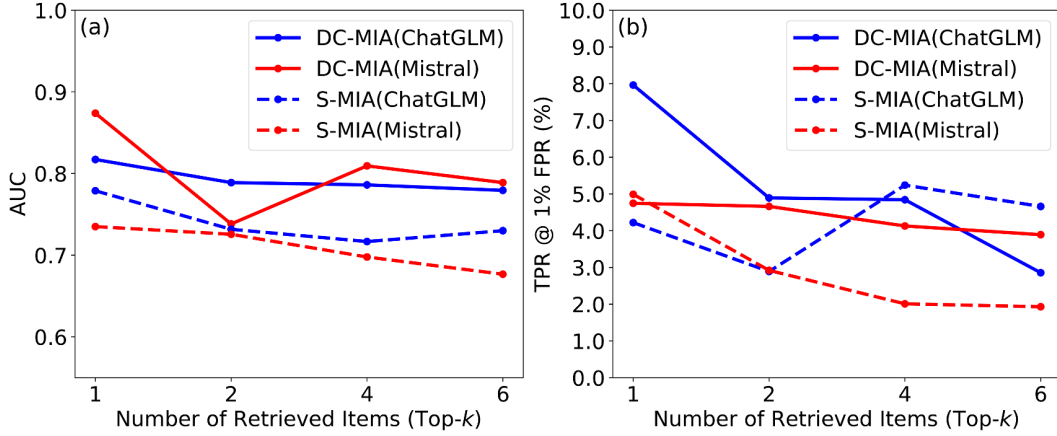


Figure 7 (Color online) Impact of the number of retrieved items (Top-k): experiments on ChatGLM and Mistral with AgNews. (a) AUC variation with respect to Top-k; (b) TPR at 1% FPR variation with respect to Top-k.

Table 2 Comparison of DC-MIA and S-MIA attack performance under different retrievers.

Retriever	HealthCareMagic				AgNews			
	AUC		TPR at 1% FPR (%)		AUC		TPR at 1% FPR (%)	
	S-MIA	DC-MIA	S-MIA	DC-MIA	S-MIA	DC-MIA	S-MIA	DC-MIA
BGE-en	0.648	0.681	1.842	2.058	0.675	0.782	2.340	3.948
BM25	0.633	0.682	1.710	2.223	0.669	0.779	2.498	3.340
Ideal	0.621	0.724	2.112	2.796	0.695	0.792	2.200	4.720

Table 3 AUC and TPR at 1% FPR for target RAGs (ChatGLM & LLaMA) attacked by reference RAG (Mistral).

Attack method	HealthCareMagic			
	AUC		TPR at 1% FPR (%)	
	LLaMA	ChatGLM	LLaMA	ChatGLM
DC-MIA	0.6517	0.6367	4.033	2.140
S-MIA	0.6294	0.5900	2.214	1.559

Different LLMs for target RAG and reference RAG. Given that the LLM parameters of the target RAG may not always be fully known, we have relaxed the assumptions of the threat model to allow reference RAGs using different LLMs to attack the target RAG. Table 3 presents the results of these attacks. The findings show that even when the reference RAG and target RAG use completely different LLM structures, DC-MIA outperforms the baseline methods. For example, when a Mistral-based reference RAGs attacks an LLaMA-based target RAG, DC-MIA achieves an AUC of 0.6517 and a TPR of 4.033% at 1% FPR, compared to the baseline's 0.6294 AUC and 2.214% TPR. These results suggest that DC-MIA's attack performance is not heavily reliant on prior knowledge of the target RAG's LLM parameters, it performs well even when the reference and target RAGs use different LLM structures. This can be attributed to the difficulty calibration process, where samples are compared based on their in-distribution and out-of-distribution states, with the key difference being their membership status. Overall, these results highlight the practical applicability of DC-MIA in scenarios where the LLM parameters of the target RAG are unknown.

7 Discussion

7.1 Similarity without RAG

Due to similarity measuring response difficulty rather than direct membership status, we discuss in this section the trend of similarity if a member lacks the assistance of the RAG retriever. When members lose membership status (i.e., directly using LLM answer members and no longer leveraging the context), the difference in similarity will respond to the potential sample diversity. In other words, we demonstrate the existence of sample diversity by comparison.

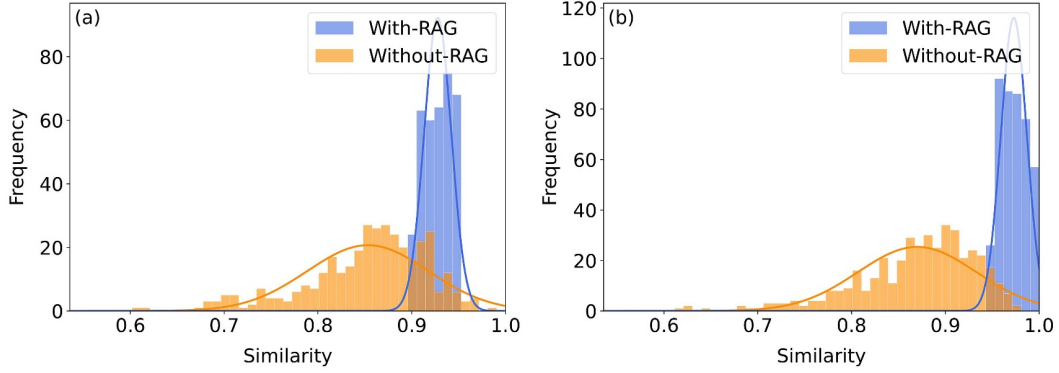


Figure 8 (Color online) Difference in similarity distribution between overlapping (a) and non-overlapping (b) samples in Figure 1(a), comparing cases with and without RAG.

Table 4 Comparison of attack performance using various ROUGE scores and similarity.

LLaMA with AgNews					
Metric	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE-Lsum	Similarity
AUC	0.803	0.834	0.788	0.809	0.630
TPR at 1% FPR (%)	4.569	1.256	3.643	4.597	1.643
Mistral with AgNews					
Metric	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE-Lsum	Similarity
AUC	0.891	0.911	0.887	0.887	0.809
TPR at 1% FPR (%)	1.437	10.610	6.313	6.321	2.011

In Figure 8, we select the members with high similarity scores in the range of $[0.9, 1.0]$ and evaluate them using LLM, both with and without RAG. Figure 8(a) shows the members whose similarity falls within the interval $[0.9, 0.95]$ in Figure 1(a) (overlapping samples). When these members in the overlapping region do not use context, their similarity scores decrease and spread across the range $[0.7, 0.95]$, while the non-members remain within $[0.9, 0.95]$. This demonstrates that when the influence of membership is removed from the similarity measure, the similarity of these samples in the without-RAG case shows a significant decline and exhibits a different distribution compared to the with-RAG case.

Furthermore, Figure 8(b) illustrates the members whose similarity falls within the interval $[0.95, 1.0]$ in Figure 1(a) (mostly non-overlapping, with a small overlap, as the sample size for fully non-overlapping parts is too small for meaningful distribution analysis). These members exhibit particularly high similarities in the with-RAG case. However, the similarity distribution for the without-RAG case remains in the range $[0.7, 0.95]$, which is similar to the distribution in Figure 8(a), indicating that the effect of sample diversity on similarity is inherent. Therefore, membership cannot be directly represented by similarity scores, as is commonly done in existing works. Instead, it should be calibrated using our method proposed in this paper.

7.2 Beyond similarity

Here, we clarify that our DC-MIA is not limited to using only similarity as the raw score to be calibrated. Specifically, in addition to the similarity score used in [14], we also experimented with other metrics, including ROUGE-1, ROUGE-2, ROUGE-L, and ROUGE-Lsum, which belong to the recall-oriented understudy for gisting evaluation (ROUGE) [59]. The results are shown in Table 4. It can be observed that some metrics outperform similarity, while others show weaker performance. Furthermore, the performance of the same metric varies significantly across different RAG systems. For example, on LLaMA with AgNews, ROUGE-2 achieved the lowest TPR at 1% FPR (1.256%), while on Mistral with AgNews, ROUGE-2 achieved the highest TPR at 1% FPR (10.610%). Therefore, we suggest that a variety of metrics should be explored for different target RAG systems in order to achieve optimal attack performance.

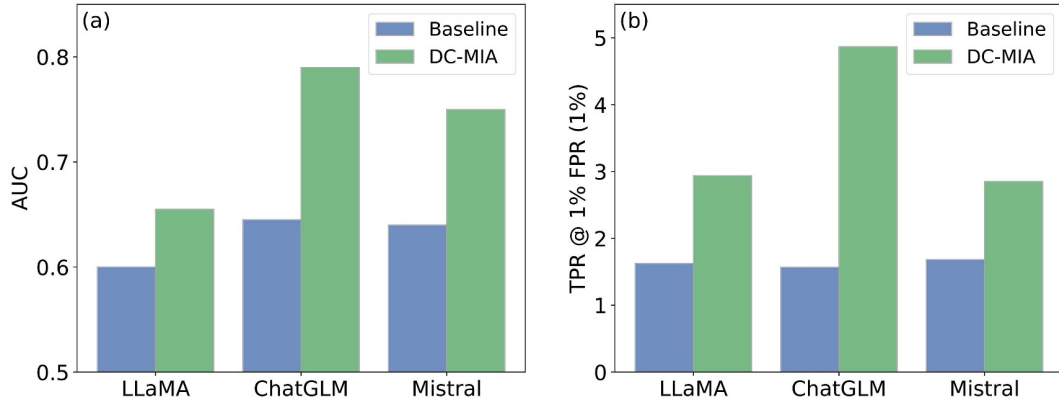


Figure 9 (Color online) Comparison of the attack performance of DC-MIA and S-MIA on three RAG systems built with LLaMA, ChatGLM, and Mistral using the NaturalQuestions dataset. (a) AUC comparison; (b) TPR at 1% FPR comparison.

7.3 Practical scenarios

To validate the effectiveness of DC-MIA attacks in real-world scenarios, we conducted experiments on three RAG systems using LLMs (LLaMA, ChatGLM, and Mistral) with the NaturalQuestions dataset. The evaluation used the ROUGE-L metric, as described in Subsection 7.2. The results are shown in Figure 9. It can be observed that DC-MIA consistently outperforms baseline methods across all scenarios. For example, on Mistral, DC-MIA achieved a TPR of over 3% at 1% FPR, while S-MIA was below 2%. Additionally, DC-MIA's AUC was 0.750, significantly higher than S-MIA's 0.640. These findings clearly demonstrate DC-MIA's superior attack capability on datasets commonly used in RAG applications. The consistent advantages in both TPR at 1% FPR and AUC further confirm DC-MIA's effectiveness and robustness in practical settings.

8 Conclusion and future work

In this study, we introduce a novel MIA for RAG systems, termed DC-MIA, which specifically addresses the impact of sample difficulty on the performance of MIAs. DC-MIA works by first classifying samples in high-similarity regions as members, and then calibrating the membership scores of samples with comparable raw similarity scores using a likelihood ratio test.

Our extensive experiments show that DC-MIA outperforms the baseline in nearly all cases, particularly in terms of TPR at low FPR. This suggests that prior studies may have underestimated privacy risks in RAGs, as they did not account for the varying levels of difficulty across different samples. Additionally, we conduct an in-depth analysis of key factors influencing the performance of DC-MIA.

Notably, our analysis of the number of retrieved items reveals that increasing this number reduces the effectiveness of MIAs. Despite this, the performance of these attacks was not reduced to an acceptable level. Given these findings, we recommend minimizing the use of sensitive data in RAG systems. In the future, we will investigate privacy-preserving defense mechanisms for RAG systems to ensure their security and confidentiality.

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